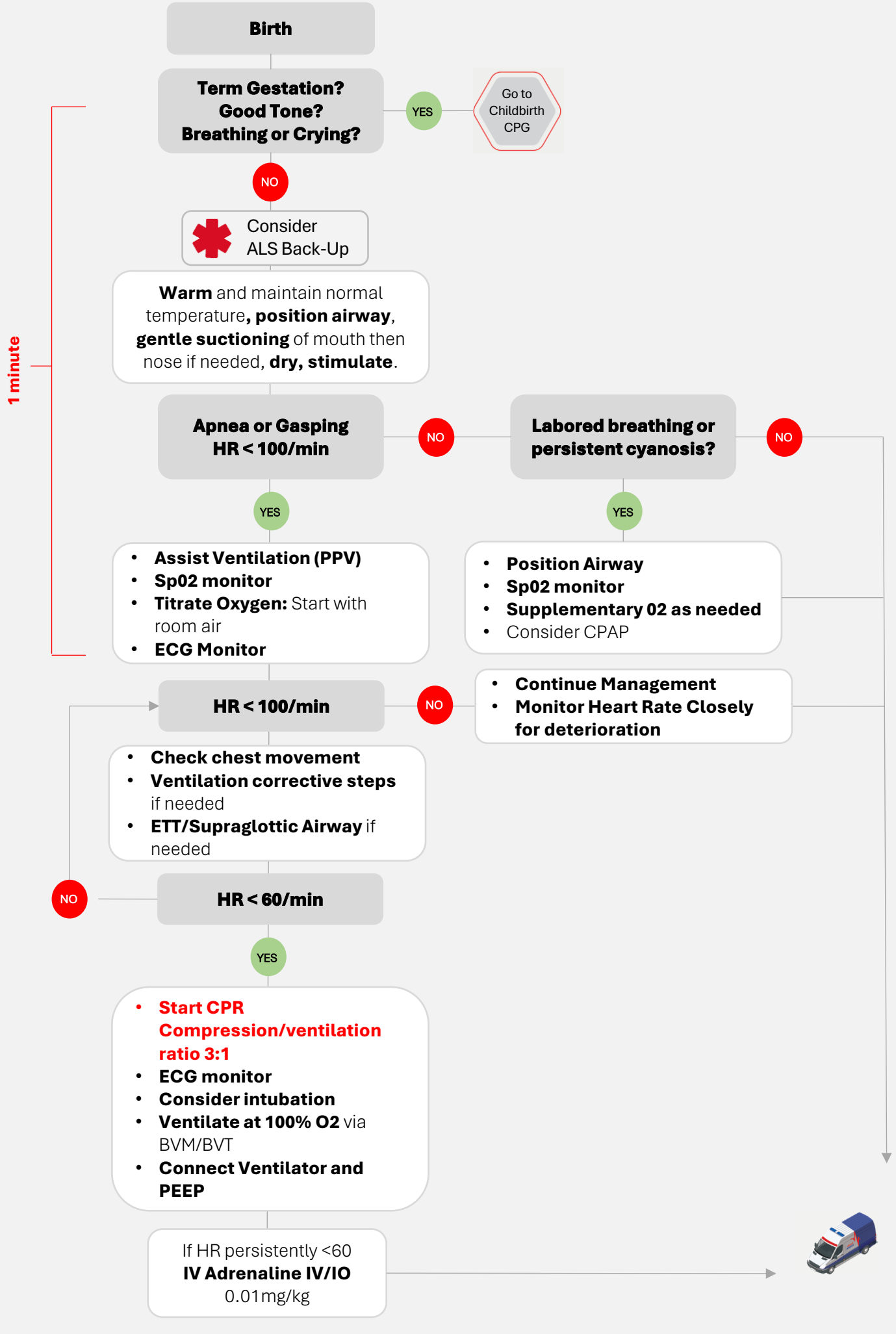
Newborn (≤4weeks) and Preterm Infant Resuscitation



Patient Care Goals

1. Provide routine care to the newborn infant.
2. Rapidly identify newly born infants requiring resuscitative efforts.
3. Establish spontaneous breathing and a stable heart rate in newborns and preterm infants.
4. Maintain normothermia and provide adequate oxygenation while minimizing complications such as hyperoxia or hypoxia.
5. High-quality chest compressions/CPR with minimal interruptions from recognition of cardiac arrest until confirmation of ROSC or termination of care.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Newborns (< 4 weeks) and preterm infants requiring resuscitation due to:
  - Apnea or gasping respiration.
  - Heart rate <100 bpm at 1 minute post-delivery.
  - Cyanosis or pale appearance.
  - Poor muscle tone or lack of crying.

Exclusion Criteria

- Infants >4 weeks of age (See Pediatric Resuscitation CPG).

Key Patient Assessment Points

1. **Assessment**
   - **Patient History**
     - Date and time of birth
     - Prenatal history (prenatal care, substance abuse, multiple gestation, maternal illness.
     - Birth history (maternal fever, presence of meconium, maternal bleeding, difficult delivery – e.g. shoulder dystocia, prolapsed cord or nuchal core, breech)
     - Estimated gestational age (may be based on last menstrual period)
2. **Physical Examination & Assessment**
   - Respiratory rate and effort (strong, weak, or absent; regular or irregular)
   - Signs of respiratory distress (grunting, nasal flaring, retractions, gasping, apnea)
   - Pulse oximetry should be considered if resuscitative efforts are initiated or if supplemental oxygen is administered.
   - Heart rate (fast, slow, or absent)
     - Precordium, umbilical stump, or brachial pulse may be used.
     - Auscultation is preferred since palpation of the umbilical stump is less accurate.
   - Muscle tone (poor or strong)
   - Color/Appearance (central cyanosis, acrocyanosis, pallor, normal)
   - APGAR score at 1 minute and 5-minute post full delivery of infant.
   - **APGAR Score Table:**

	Sign	2	1	0
A	**Appearance** (skin color)	Normal over entire body	Normal except extremities	Cyanotic or pale all over
P	**Pulse**	>100 bpm	<100 bpm	Absent
G	**Grimace** (reflex irritability)	Sneezes, coughs or vigorous cry	Grimaces	No response
A	**Activity** (Muscle tone)	Active	Arms and legs flexed	Absent
R	**Respirations**	Good, crying	Gasping, irregular	Absent

3. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.
        - All patients who required ventilatory support or chest compressions after delivery are Red Priority (even if they demonstrated improvement).

Patient Management & Interventions

1. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
2. **Immediately After Delivery**
    - Assess for tone, breathing or crying.
        - If tone, breathing or crying is adequate, no resuscitation is necessary.
            - Go to Childbirth CPG.
    - **If weak cry, signs of respiratory distress, poor tone, or preterm gestation:**
        - **Position airway** (sniffing position) and clear airway as needed.
        - If signs of respiratory distress with airway obstruction, suction mouth then nose; routine suctioning is not recommended.
        - **Dry, warm, and stimulate**
            - Wrap infant in dry towel/thermal blanket or clean plastic bag up to the neck, to keep infant as warm as possible; keep head covered if possible.
3. **Reassess Breathing and Heart Rate**
    - **If Laboured Breathing or Persistent Cyanosis:**
        - Attach Sp02 monitor
        - Consider Supplementary 02 (start with low-Fi02 initially)
    - **Apnea or Gasping:**
        - Position Airway
            - Consider Airway Adjuncts (OPA/NPA/Intubation)
        - Assist Ventilation (PPV) at 40-60 bpm.
            - Do not add oxygen initially.
        - Attach Sp02 monitor
        - Attach ECG Monitor
    - **Consider Cutting Cord:**
        - If the patient does not rapidly improve with interventions, or requires immediate resuscitation:
        - Clamp the cord in three places (10cm, 15cm, 20cm) and cut between the 2 clamps furthest away from the newborn.
4. **If Heart Rate < 100**
    - Initiate BVM ventilation with room air at 40-60 breaths per minute for 90 seconds with room air.
        - The primary indicator of effective ventilation is improvement in heart rate.
        - Rates and volumes of ventilation can be variable, only use the minimum necessary rate and volume to achieve chest rise and a change in heart rate.
    - Evaluate heart rate every 30 seconds
        - If no improvement after 90 seconds, change oxygen delivery to 100% Fi02 until the heart rate normalizes
4. **If Heart Rate < 60**
    - Initiate chest compressions:
        - Two-thumb-encircling-hands technique is preferred.
        - 3:1 chest compression – ventilation ratio.
    - Consider endotracheal intubation or supraglottic airway.
    - Establish IV or IO Access.
    - Administer Epinephrine:
        - Adrenaline: 0.01 mg/kg IV/IO every 3-5 minutes if the heart rate remains less than 60bpm.
        - IV Fluid Administration
    - Administer 20 mL/kg normal saline IV/IO for signs of shock or post-resuscitative care.

- Check blood glucose.

Patient Safety Considerations

- Hypothermia is common in newborns and worsens outcomes of nearly all post-natal complications.
  - Ensure heat retention by drying the infant thoroughly, covering the head, and wrapping the baby in dry cloth.
  - Kangaroo care advisable where no resuscitative care is needed is an effective warming technique (i.e. placing the infant skin-to-skin directly against the mother's chest and wrapping them together)
  - Newborn infants are prone to hypothermia which may lead to hypoglycemia, hypoxia, and lethargy. Aggressive warming techniques should be initiated, including drying, swaddling, and warm blankets covering body and head.
- Preterm infants (<23 weeks) may receive comfort care based on parental wishes; resuscitation is not routinely recommended.

Clinical Insights & Additional Guidance

- Most newborns require only drying, warming, and stimulating to help them transition from fetal respiration to newborn respiration.
- **Resuscitation Strategy**
  - The resuscitation sequence can be remembered as: Dry, Warm, and Stimulate – Ventilate – Evaluate – and Resuscitate.
- **Placental Delivery**
  - If the placenta is delivered, place it in a biohazard bag and transport it with the patient to the hospital.
- **Newborn Pulse Oximetry:**
  - Measuring the pulse oximetry on the right hand provides the most accurate oxygen saturation in infants that are transitioning from fetal to normal circulation.
  - At 60 seconds, 60% is the target with an increase of 5% every minute until 5 minutes of life when pulse oximetry is 80-85%.
  - Projected Pulse Oximetry in Infants Over Time:

Time Since Birth	Projected Increase in Pulse Oximeter Over Time
1 minute	60-65%
2 minutes	65-70%
3 minutes	70-75%
4 minutes	75-80%
5 minutes	80-85%
10 minutes	85-90%

- **Hypoxia & Oxygen Administration**
  - Both hypoxia and excess oxygen administration can result in harm to the infant. If prolonged oxygen use is required, titrate to maintain an Sp02 of 85%-95%
  - Increase in heart rate is the most reliable indicator of effective resuscitative efforts.
- **Pertinent Assessment Findings**
  - If there is any doubt in gestational age in the field, resuscitation efforts should be initiated.
  - Acrocyanosis, a blue discoloration of the distal extremities, is a common finding in the newly born infant transitioning to extrauterine life. This must be differentiated from central cyanosis.
- **Pre-Term Infant Resuscitation**
  - Avoid routine resuscitation <22 weeks unless parental wishes and clinical judgement indicate otherwise.

Documentation

- Identification: Infant's gender, estimated gestational age, and time of birth.
- Mechanism / Medical Complaint: Symptoms such as apnea, gasping, poor muscle tone, cyanosis.
- Injuries / Information:
  - Prenatal history (e.g., prenatal care, maternal substance abuse, multiple gestation, maternal illness).
  - Birth history (e.g., maternal fever, meconium presence, maternal bleeding, difficult delivery such as shoulder dystocia, prolapsed cord, nuchal cord, or breech).
- Signs:
  - Vital signs (heart rate via precordium/umbilical stump/brachial pulse, respiratory rate, SpO_2 if monitored).
  - APGAR score at 1 and 5 minutes post-delivery (appearance, pulse, grimace, activity, respirations).
  - Physical findings (e.g., central cyanosis, acrocyanosis, pallor, grunting, nasal flaring, retractions, muscle tone).
  - Blood glucose level (if checked).
- Treatment:
  - Airway management (e.g., positioning in sniffing position, suctioning mouth then nose if obstruction present).
  - Warming and stimulation (e.g., drying, wrapping in thermal blanket/plastic bag, covering head).
  - Ventilation details
  - Airway adjuncts (e.g., OPA/NPA, endotracheal intubation, supraglottic airway).
  - IV/IO access placement and fluid administration
  - Medication administration
  - Cord management
  - Oxygen therapy
- Allergies: Not typically applicable for newborns.
- Medications: Note any maternal medications (e.g., tocolytics) relevant to the infant's condition.
- Background: Maternal chronic illnesses (e.g., diabetes, hypertension), recent hospitalization, or obstetric complications.
- Other:
  - Placenta management (e.g., placed in biohazard bag for transport if delivered).
  - Dispatch and hospital coordination notes (e.g., notification of neonatal resuscitation and receiving facility preparation).
  - Note parental wishes and clinical judgment for preterm infants <23 weeks.

Key Performance Indicators

1. Provide chest compressions and ventilation within 30 seconds of the heart rate falling below 60bpm.

References

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4. Katheria, A.C. et al. (2019) 'Umbilical Cord Milking vs Delayed Cord Clamping in Preterm Infants: A Systematic Review and Meta-Analysis', *American Journal of Obstetrics & Gynecology*, 220(6), pp. 545–556.
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